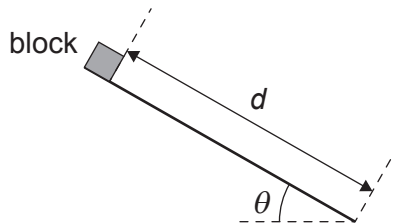


PAPER 1 / QUESTION 134

8825-6707 / original question 5

- 134.** A block of mass  $m$  is released from rest and slides down a ramp of length  $d$  that makes an angle  $\theta$  with the horizontal. A constant frictional force  $F$  acts on the block.



What is the kinetic energy of the block at the bottom of the ramp?

- A.  $mgd - Fd \cos \theta$
- B.  $mgd - Fd$
- C.  $mgd \sin \theta - Fd \cos \theta$
- D.  $mgd \sin \theta - Fd$